

Helix OTA — Multi-Format Documentation Export Pipeline

Field	Value
Revision	1
Created	2026-06-07
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Status	active
Status summary	The real, tested pipeline that converts the canonical Markdown corpus to HTML + DOCX (+ PDF when a LaTeX engine is present) and renders Mermaid diagrams to SVG + PNG. Implemented in <code>scripts/export_docs.sh</code>. This document records the exact commands and the real captured output of a live run (§11.4.6/§7.1 — no claimed format unless a file was physically produced). PDF and draw.io/PlantUML/graphviz rendering are not available on the current host (no LaTeX engine; no drawio/plantuml/dot). These are SKIPPED, not faked.
Issues	HTML, DOCX, Mermaid→SVG, Mermaid→PNG verified working. PDF + native draw.io/UML render require additional tooling (documented below).
Issues summary	initial pipeline + live verification
Fixed	<code>scripts/export_docs.sh</code> written, made executable, and run against the master design doc; outputs verified on disk.
Fixed summary	Add a CI job + container (pandoc + mermaid-cli + a LaTeX engine such as tectonic, + drawio-desktop/plantuml) so every format renders reproducibly; wire into the changelog/release export (§5).
Continuation	

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§1. Goal & Constitution basis

HelixConstitution §11.4.65 mandates that every Markdown document be exportable to **PDF, HTML, DOCX**, and that diagrams be available as **Mermaid, draw.io, SVG, UML, PNG**. The canonical source of truth is **Markdown + Mermaid**; all other formats are *generated* from it (and therefore git-ignored per §11.4.30, regenerable on demand). Per §7.1/§11.4.6 this document claims a format works **only** when a file was physically produced; everything else is marked SKIPPED with the reason.

§2. Tooling — verified host capabilities

Probed live on the build host (2026-06-07):

Tool	Status	Used for
pandoc	✓ FOUND (/opt/homebrew/bin/pandoc)	Markdown → HTML, DOCX, PDF
mmdc (mermaid-cli)	✓ FOUND (/opt/homebrew/bin/mmdc)	Mermaid → SVG, PNG
npx, node v22, java 17	✓ FOUND	tooling/runtime
LaTeX engine (tectonic/xelatex/pdflatex/lualatex/wkhtmltopdf)	✗ MISSING	(blocks PDF)
drawio	✗ MISSING	(blocks draw.io→PNG render)
plantuml	✗ MISSING	(blocks UML render)
dot (graphviz)	✗ MISSING	(blocks graphviz render)

§3. The script

`scripts/export_docs.sh <file.md | dir> [out_dir]` (defaults to docs/research/main_specs). It:

1. Probes capabilities at runtime; logs what it can/can't do.
2. For each .md (excluding _exports/): produces *.html and *.docx via pandoc; attempts *.pdf only if a PDF engine exists (else SKIP + warn).

3. Extracts each fenced ``mermaid block to a numbered.*diagram-NN.mmd* and renders.svg+*.png via mmdc`.
4. Degrades gracefully — a missing renderer is logged, never fatal.
5. Prints a final tally (html= docx= pdf= mermaid_blocks= svg= png=).

Outputs land in a sibling `_exports/` directory (git-ignored).

§4. Live run — real captured evidence

Command:

```
$ ./scripts/export_docs.sh docs/research/main_specs/00-master/
2026-06-07-helix-ota-design.md
```

Real stdout:

```
[export] capabilities: pandoc=1 mmdc=1 pdf_engine=none
[export][WARN] no LaTeX/PDF engine — PDF will be SKIPPED (HTML/
DOCX unaffected)
[export] inputs: 1 markdown file(s)
[export] DONE — html=1 docx=1 pdf=0 mermaid_blocks=1 svg=1 png=1
```

Resulting files (`ls -la ../_exports/`):

```
2026-06-07-helix-ota-design.diagram-01.mmd    1081 B
2026-06-07-helix-ota-design.diagram-01.png    33477 B
2026-06-07-helix-ota-design.diagram-01.svg    27820 B
2026-06-07-helix-ota-design.docx             19249 B
2026-06-07-helix-ota-design.html              22650 B
```

This is rock-solid physical evidence: HTML, DOCX, Mermaid-SVG and Mermaid-PNG were all produced (non-zero, valid sizes). PDF was correctly SKIPPED — not faked.

§5. Format coverage matrix (honest status)

Required format (§11.4.65)	Status on this host	Mechanism
HTML	✅ verified	pandoc
DOCX	✅ verified	pandoc
PDF	⏸ SKIPPED — needs LaTeX engine	pandoc + tectonic/ xelatex
Mermaid (source)	✅ canonical	authored inline
SVG	✅ verified	mmdc
PNG	✅ verified	mmdc
draw.io	⚠ source-only (no local renderer)	emit .drawio XML; render needs drawio- desktop

Required format (§11.4.65)	Status on this host	Mechanism
UML	⚠ source-only (no local renderer)	emit .puml; render needs plantuml

§6. Making PDF + draw.io + UML render

To reach 100% of §11.4.65 reproducibly, the export container (built on the `containers` submodule) must add: a LaTeX engine (tectonic recommended — single binary, no texlive sprawl), `drawio-desktop` (headless `--export`), `plantuml` (+ `graphviz`). The script already calls these conditionally, so adding the binaries flips the SKIPPED rows to verified with no script change. This containerization is tracked as a 1.0.0-MVP deployment task.